# Case 36: Slope-factor Rothermel sweep

## 1 Verification purpose

This case verifies the scalar Rothermel slope response for FBFM2, FBFM9, GR2, TU5, and TL8 over 0, 10, 20, 30, 50, 75, and 100% rise over run. It tests slope-raster units, conversion to $\tan^2\phi$, fuel-dependent packing response, and ELMFIRE's slope-factor cap without wind. The verification question is whether all 35 simulation conditions reproduce independently calculated intensity and slope-enhanced rate of spread (ROS) within the stated tolerances.

## 2 Mathematical formulation and numerical implementation

The no-wind response is

$$R = R_0(1 + \phi_s), \qquad \phi_s = 5.275\,\beta^{-0.3}\tan^2\phi. \tag{1}$$

Here reported slope percent is rise/run times 100 and the raster value is the angle $\phi = \tan^{-1}(s/100)$ in degrees. The formulation follows Andrews (2018), RMRS-GTR-371. The inspected ELMFIRE source additionally caps $\phi_s$ at the maximum wind factor derived from its wind limit; the reference calculation reproduces and reports that implementation policy.

## 3 Derivation of expected behavior

Before the cap, plotting $\phi_s$ against $\tan\phi$ on logarithmic axes has slope two. For each fuel and raster angle, the local reference calculation calculates $R_0$, the uncapped factor, the cap, and final $R$. Because the initial level set is planar and aligned with the upslope direction, the output normal speed equals this scalar head-spread prediction.

The slope-only reference uses

$$\phi_s = 5.275\,\beta^{-0.3}\tan^2\theta, \qquad R = R_0(1 + \phi_s).$$

As an explicit generated substitution, the requested 20 percent slope is stored as the integer raster angle $\theta = 11$ degrees ($100\tan 11^\circ = 19.4380$ percent). For Fuel Model 2, $\beta = 0.00575$, $R_0 = 0.8237955$ m min$^{-1}$, and

$$\phi_s = 5.275(0.00575)^{-0.3}\tan^2(11^\circ) = 0.936756,$$
$$R_{\text{ref}} = 0.8237955(1 + 0.936756) = 1.595491 \text{ m min}^{-1}.$$

Reaction intensity is unchanged by slope; here it is $675.184$ kW m$^{-2}$. Every other generated fuel/slope pair uses the same equations with its stored integer angle and fuel packing ratio. Simulation condition errors are $e_R = |R_{\text{max}} - R_{\text{ref}}|/\max(|R_{\text{ref}}|, 10^{-8})$ and analogously $e_I$. From $t = a + bx_{\text{surface}}$, $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$. The aggregate substitutions are $\max e_R \le 0.02$, $\max e_I \le 0.01$, $\max e_{\text{TOA}} \le 0.05$, and $\min R^2 \ge 0.995$.

## 4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS, 0.01 for reaction intensity, and 0.05 for time of arrival (TOA)-derived ROS; TOA regression requires $R^2 \ge 0.995$. The direct and reference calculation ROS

values are slope-surface speeds. Accordingly, the TOA regression converts map easting to distance along the sloping surface using $x_s = x_0 + (x - x_0)/\cos\theta$ before fitting speed. Pointwise analytical errors are used instead of a trend-only test because degrees-versus-percent mistakes can preserve monotonicity. All simulation conditions and output fields must exist.

## 5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE's supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive reference spread rate, the prescribed and maximum timesteps use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured Courant–Friedrichs–Lewy (CFL) limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE's stalled-front branch and corrupt the terminal timestamp. A zero-ROS simulation condition retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required simulation condition must reach the prescribed stop time in its generated namelist; a direct-only simulation condition may terminate early because lack of propagation is part of the state being tested. Each 1400 m square domain uses 10 m cells and a finite ignition strip with a planar east face in the interior analysis corridor. The strip does not touch the lateral grid boundaries. Wind is zero, dead moisture is fixed at 5/7/9%, and live moisture is 90%. Aspect is chosen so upslope is aligned with propagation. Slope inputs are generated as explicit integer degrees after applying the positive-value equivalent of Fortran NINT; the expected response and slope-surface coordinate conversion use that same generated angle. ROS should rise quadratically in $\tan\phi$ before any source-defined plateau.

| Parameter | Configured value | Units | Role |
|---|---|---|---|
| Domain/grid | 1400 m square; 10 | m/cell | Planar upslope propagation |
| Fuel sweep | FBFM2, FBFM9, GR2, TU5, and TL8 | code/name | Fuel-dependent packing response |
| Slope sweep | 0, 10, 20, 30, 50, 75, and 100% | percent rise/run | Independent terrain target |
| Wind/moisture | calm; dead 5/7/9, live 90/90 | mph / percent | Isolates slope factor |
| Timing | generated whole-second step and exact-multiple stop | s | CFL-safe terminal evidence |

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

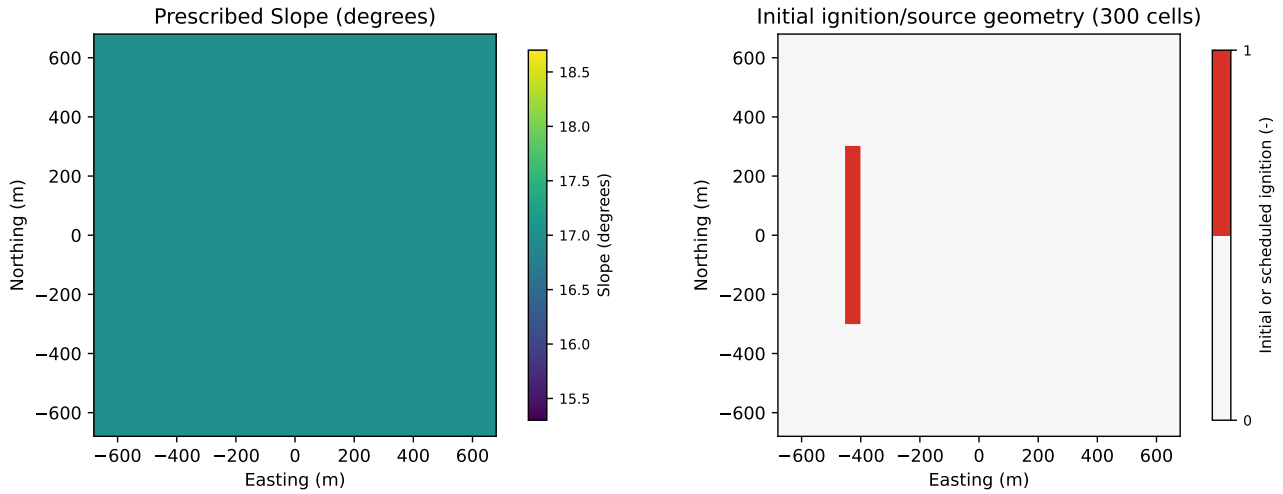



Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

## 6 Actual simulation results

The available results cover 35 of 35 simulation conditions and has state **PASS**.

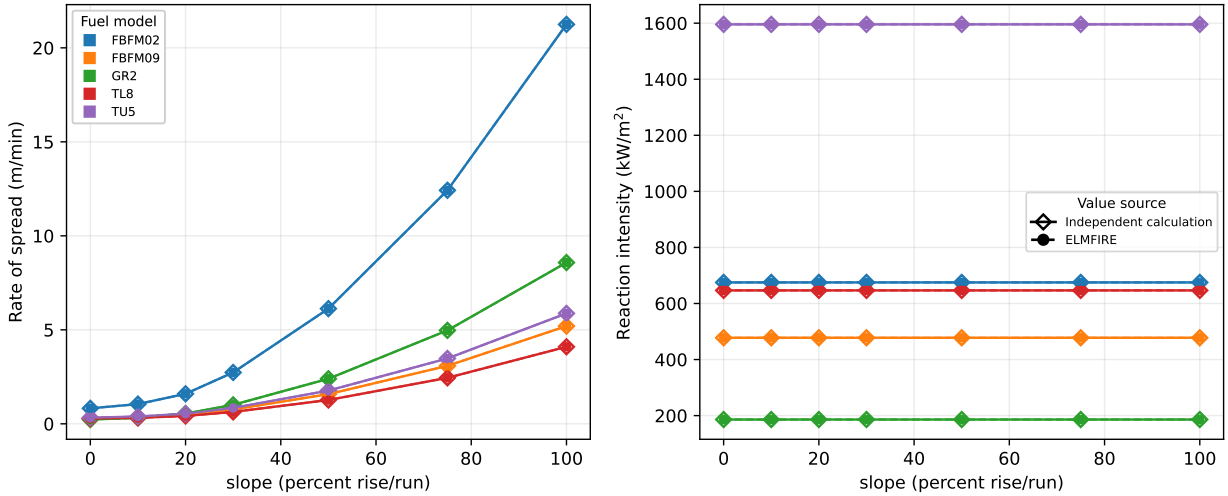



Figure 2: Independent calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines). Each fuel-model pair uses the same color.

## 7 Calculated metrics and verification decision

| Metric | Limit | Calculated | Status |
|---|---|---|---|
| maximum direct-ROS relative error | $\leq 0.02$ | 5.4278e-07 fraction | PASS |
| maximum reaction-intensity relative error | $\leq 0.01$ | 1.98901e-07 fraction | PASS |
| maximum TOA-ROS relative error | $\leq 0.05$ | 0.0265981 fraction | PASS |
| minimum TOA regression R2 | $\geq 0.995$ | 0.999961 dimensionless | PASS |

The overall decision is **PASS**.

## 8 Technical interpretation

Large errors growing with slope indicate a units or tangent error. Fuel-dependent errors implicate packing ratio or table selection. A premature or absent plateau indicates changed cap semantics. An accurate direct field but biased TOA indicates propagation or coordinate handling.

## 9 Scope and limitations

This test covers uniform positive slopes aligned with the front. It does not verify vector competition between wind and slope, complex terrain, or experimental fire behavior; CASE40 covers the directional coupling.